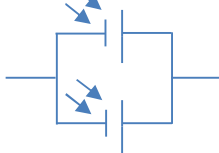
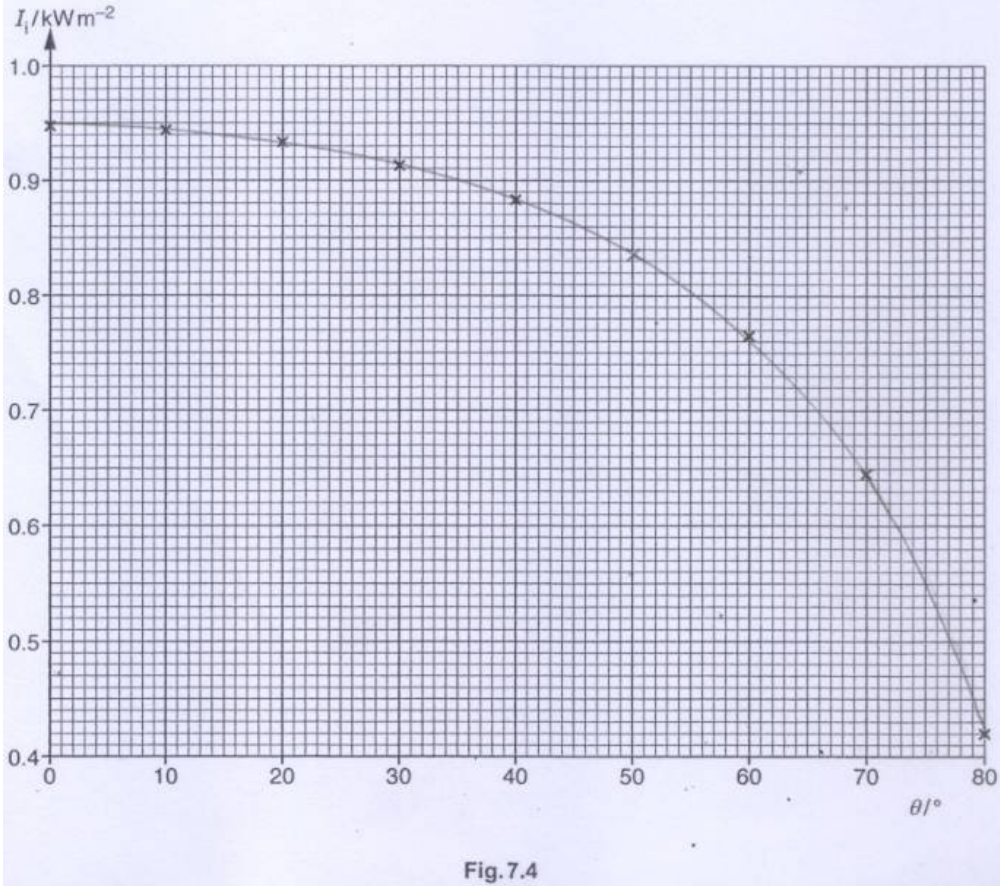


7(a)(i)	With a glide ratio of 40, a 1 m vertical drop results in a 40 m horizontal distance travelled. From cruising altitude 8500 m to night time altitude 1500 m, there is a vertical drop of 7000 m. Horizontal distance travelled = $(40)(7000) = 2.8 \times 10^5$ m.
7(a)(ii)	1. Large wing span (71.9 m) 2. Small mass (2300 kg)
7(a)(iii)	When the aircraft is in horizontal flight at constant speed, the vertically upward force acting on it (the lift) balances its weight = $(2300)(9.81) = 22563$ N The glide ratio is also the lift to drag ratio so the drag will be smaller than the lift by a factor of 40. Drag = $\frac{22563}{40} = 564.1 = 560$ N
7(b)(i)	Since kWh is the unit for energy and kg is the unit for mass, energy density having the unit kWh kg⁻¹ refers to the energy stored per unit mass of the battery.
7(b)(ii)	Each fully charged battery can store 41 kWh of energy. 4 fully charged batteries can store $(4)(41) = 164$ kWh of energy 164 kWh = $(164)(1000 \text{ W})(3600 \text{ s}) = 5.904 \times 10^8 = 5.9 \times 10^8$ J
7(b)(iii)	The energy from the batteries is used to do work against the drag force. (distance travelled)(drag force) = energy from batteries (distance travelled)(564.1) = 5.904×10^8 distance travelled = $1.047 \times 10^6 = 1.05 \times 10^6$ m
7(c)	At higher altitudes there are less clouds and pollutants to block the sunlight therefore more energy can be collected by the solar cells.
7(d)	The aircraft needs to be in continuous flight but it is unable to collect solar energy at night when it is dark. Therefore it must collect more than it uses during the day, store the excess unused energy in the batteries to be used at night.
7(e)	1. Connect the solar batteries in parallel:  2. Connect the solar batteries in series: 
7(f)(i)	The power collected by the cells is proportional to the area of the cells.
7(f)(ii)	The gradient is the power collected per unit area of the cells, the intensity of power collection.
7(f)(iii)	$I_c = \frac{73000}{40} = 243.3 = 240 \text{ W m}^{-2}$

7(g)(i)	For $\theta = 50^\circ$: $M = \frac{1}{\cos \theta} = \frac{1}{\cos 50^\circ} = 1.556$ $I_i = 1.353 \times 0.700^k$ Since $k = M^{0.678} = 1.556^{0.678} = 1.350$ and $0.700^k = 0.700^{1.350} = 0.6178$, $I_i = 1.353 \times 0.6178 = 0.8359 = 0.836$																				
7(g)(ii) 7(g)(iii)	 The graph shows the relationship between the angle of incidence θ (in degrees) and the incident intensity I_i (in kW m^{-2}). The curve starts at $(0, 1.0)$ and decreases as θ increases. The data points are as follows: <table border="1"> <thead> <tr> <th>$\theta / ^\circ$</th> <th>$I_i / \text{kW m}^{-2}$</th> </tr> </thead> <tbody> <tr><td>0</td><td>1.00</td></tr> <tr><td>10</td><td>0.98</td></tr> <tr><td>20</td><td>0.95</td></tr> <tr><td>30</td><td>0.92</td></tr> <tr><td>40</td><td>0.88</td></tr> <tr><td>50</td><td>0.83</td></tr> <tr><td>60</td><td>0.77</td></tr> <tr><td>70</td><td>0.68</td></tr> <tr><td>80</td><td>0.40</td></tr> </tbody> </table> Fig. 7.4	$\theta / ^\circ$	$I_i / \text{kW m}^{-2}$	0	1.00	10	0.98	20	0.95	30	0.92	40	0.88	50	0.83	60	0.77	70	0.68	80	0.40
$\theta / ^\circ$	$I_i / \text{kW m}^{-2}$																				
0	1.00																				
10	0.98																				
20	0.95																				
30	0.92																				
40	0.88																				
50	0.83																				
60	0.77																				
70	0.68																				
80	0.40																				
7(g)(iv)	For $\theta = 36^\circ$: incident intensity = 0.895 kW m^{-2}																				
7(g)(v)	For $\theta = 36^\circ$: Collected intensity = $243.3 \text{ W m}^{-2} = 0.2433 \text{ kW m}^{-2}$ Incident intensity = 0.895 kW m^{-2} $\text{Efficiency} = \frac{\text{collected intensity}}{\text{incident intensity}} = \frac{0.2433}{0.895} \times 100\% = 27.2\%$																				
7(h)	1. The energy density of solar cells is still low therefore the solar cells take up a significant portion of the plane's weight. This limits the total weight of passengers and cargo that the solar powered plane can carry. 2. This solar powered plane cannot be scaled up in size to carry the load of jet fuel powered planes. To increase the load it can carry, the big increase in the area of solar cells and the corresponding big increase in the wingspan will make the solar powered plane too big and impractical.																				